RPMGLOBAL

AMT – Strategy Definition

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. This guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note
A note for participants about the task.



Important Reminder
An important reminder for the participant about the task.



Learning Check
Check your understanding on the workshop and ask any related questions.

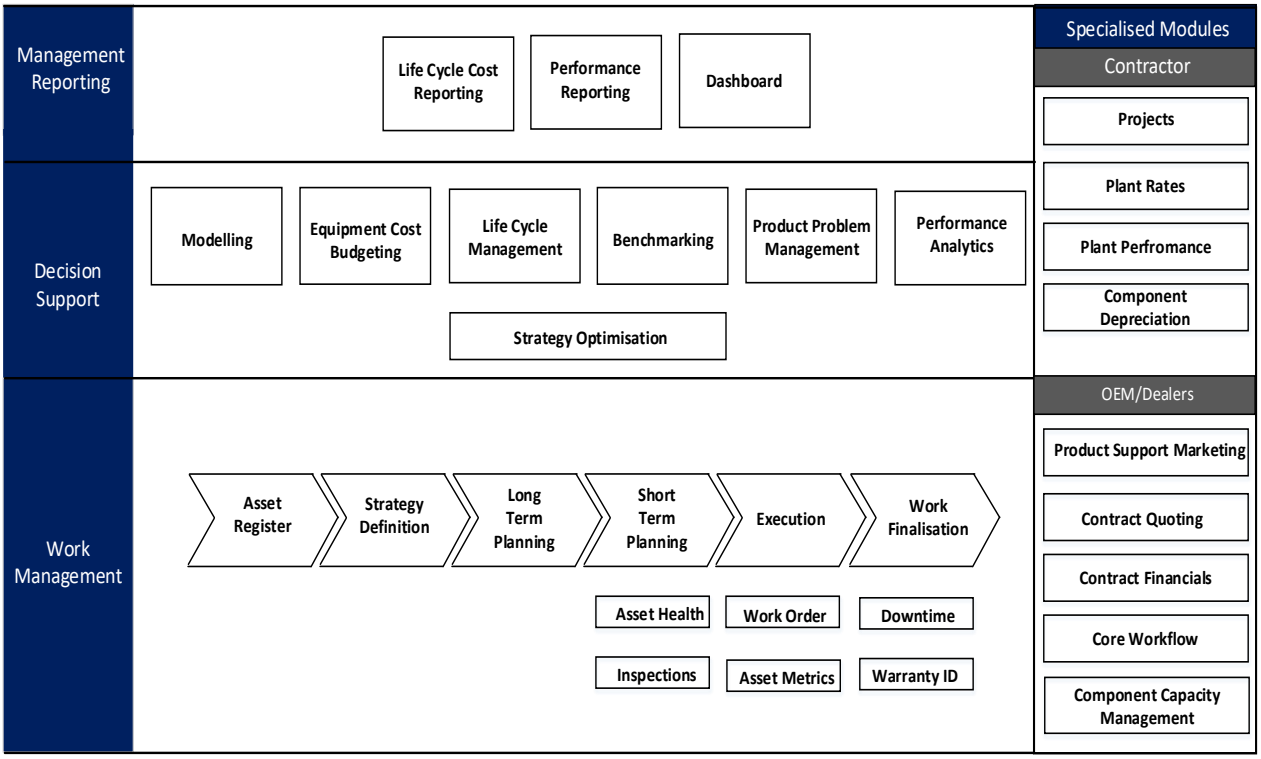
Course Outline

Pre-requisites	Getting Around AMT, AMT Key Concepts
Target Audience	Maintenance Manager, Strategist, Reliability Engineer, Contract Manager, Sales Rep. (OEM)
Nominal Duration	
Course Objectives	The objectives of Strategy Definition are: • To optimise equipment performance and minimise operating costs. • To drive continuous improvement. • To ensure AMT projections are accurate.
Learning Outcomes	At the end of the course, participants will have learned to: • Create a maintenance strategy from first principles. ○ Decide asset base operating parameters. ○ Consider asset capability. ○ Define application severity. ○ Define asset structure. • Define a maintenance strategy. ○ Determine task scheduling. ○ Define repair strategy and resource requirements. ○ Specify task instructions and documentation. ○ Understand different cost source options. ○ Link standard jobs to strategy tasks. • Creating Strategy Tasks. ○ Create Strategy Tasks ○ Create Projections ○ Override Projections • Completing Strategy Tasks. • Strategy Tasks Adjustments Factors • Editing Strategy tasks ○ Edit Strategy Tasks

	○ Forecast Consumables ○ Factor Risk • Validate Strategy Tasks ○ Create additional Non-planning Strategy Tasks ○ Set frequency of Non-planning Strategy Tasks ○ Create/Update Unassigned Strategy Tasks • Review and validate strategy. ○ Editing Multiple Strategy Tasks ○ Escalation
Assessment Strategy	The completion of workshop tasks
AMT version	8.26

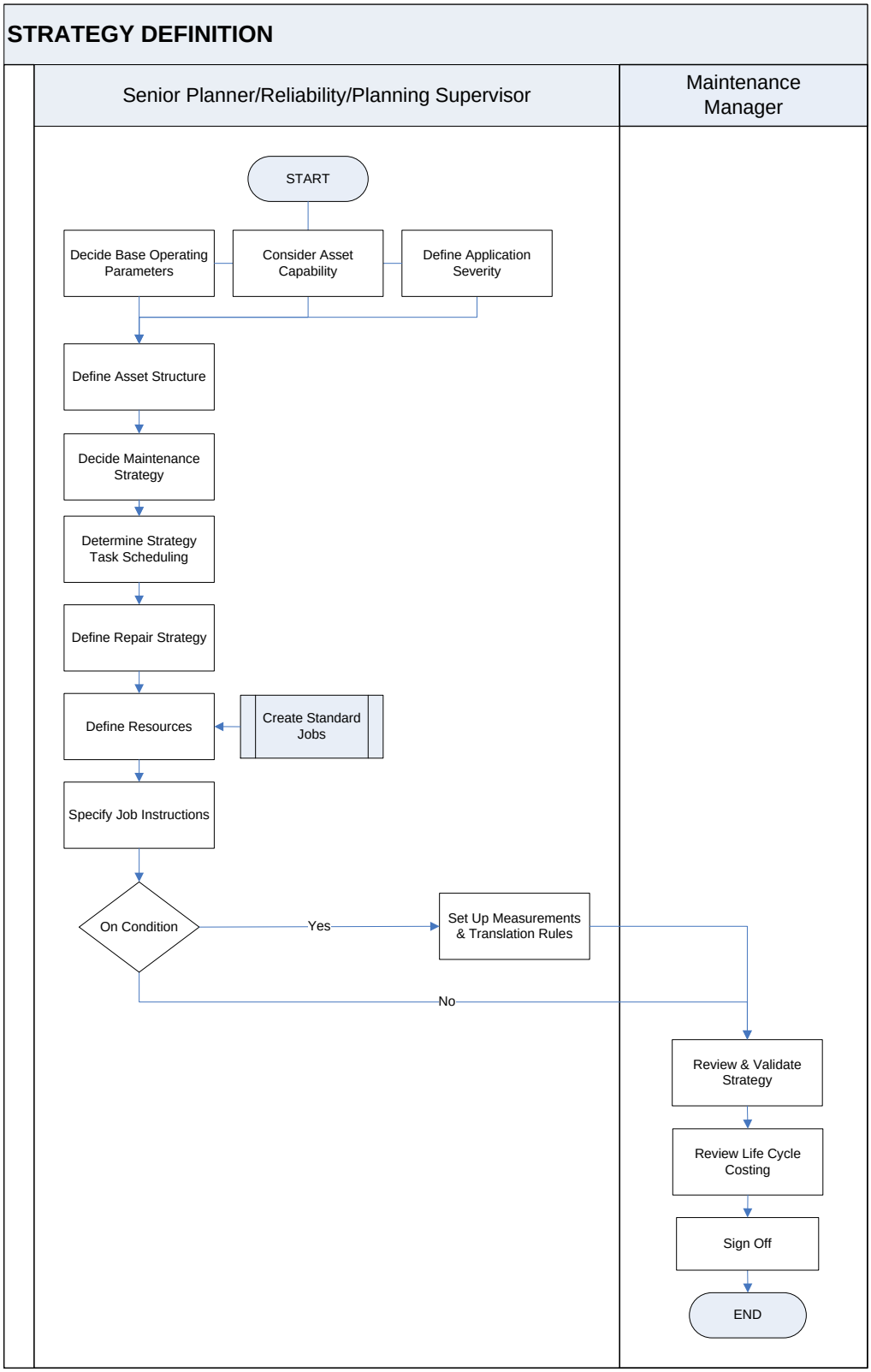
Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below. The functionality covered by this training manual is identified by the **Strategy Definition** section in the diagram.



Processes

Process Diagram



Process Inputs

The following inputs are required for this process:

- Expected equipment life, utilisation and performance requirements.
- Expected equipment application.
- Equipment design.
- Equipment function.
- Equipment and component failure modes.
- Consequences of failures.

Process Outputs

The following outputs can be expected from this process:

- Documented maintenance strategy.
- Accurate productivity, cost and resource projections.

Training Workshops

1 Decide Base Operating Parameters

Exercise – Decide Base Operating Parameters

Decide and document the base operating parameters - target life, criticality, required performance levels, incremental productivity - for (i) a mining haul truck and (ii) a loader.



This is not an exercise in AMT – use your pen and paper.

Notes

2 Consider Asset Capability

Exercise – Consider Asset Capability

From an engineering perspective, it's important to understand the capability of the asset.

- Is the asset capable of meeting the required level of performance?
- What are the application and operating factors that can influence the performance of the asset?

For example, it may be important to consider the impact of poor haul roads, overloading, or very steep haul roads on the performance of a haul truck, and which components are impacted by which factors.



This is not an exercise in AMT – use your pen and paper.

Notes

3 Define Application Severity

Procedure – Define Application Severity

Perform the following steps to review the usage profile:

1. Under **Setup & Admin** > **Usage Profiles**, double-click **Usage Profile Manager** (or search **Usage Profile Manager**).

2. To add a usage profile, click **Add New Usage Profile**.

    - To view existing projections, click **Apply** and double-click one of the results to view the details of the **Usage Profile**.

3. Enter a **Name** for the usage profile and click **Add Usage Step** to enter one or more usage steps.



Each usage step can be configured to have different usage rates. This feature is helpful if you want to create a cyclical usage profile or a profile that changes over time.

4. Click **Add New UOM** to add a unit of measure to the Parent UOM – Calendar Year.

    - Enter the **Usage per parent UOM** and select the **Unit of Measure**. You can have multiple **Parent UOMs** within a profile.

5. Click **Refresh**.



Steps in the usage profile cannot result in the equipment forecasting greater than 30 years.

6. A usage profile can be linked to an equipment in the **Term** tab of the **Equipment Editor**.

4 Define Asset Structure

Exercise – Define an Asset Structure

1. Consider a mining haul truck.
2. List the three components with the highest expected cost and downtime (also write down the suggested frequency).
3. List key systems (for example, electrical, undercarriage).
4. List a recommended PM schedule for the type of haul truck selected.
5. Decide on a valid provision amount for unscheduled maintenance for the key systems.
6. List any safety requirements (such as regular safety inspections).



This is not an exercise in AMT – use your pen and paper.

Notes

5 Decide Maintenance Strategy

Exercise – Decide Maintenance Strategy

1. Consider the three components identified in the exercise - Define an Asset Structure.
2. Decide the maintenance strategy for each of the three components.
3. Write down the reasons for selecting each maintenance strategy.



This is not an exercise in AMT – use your pen and paper.

Notes

6 Determine Task Scheduling

Exercise – Determine Task Scheduling

1. For each of the three components identified in the previous exercise – **Define an Asset Structure**, decide:

    - Date-based or usage-based
    - First occurrence
    - Frequency
    - Last occurrence

2. For the PM schedule identified in Exercise 4.4, decide:

    - Scheduling method
    - Last performed or last scheduled
    - Date-based or usage-based
    - First occurrence
    - Frequency



This is not an exercise in AMT – use your pen and paper.

Notes

7 Define Repair Strategy and Resources

Exercise – Define Repair Strategy and Resources

1. Define the repair strategy for each of the three components identified in the exercise - Define an Asset Structure.

2. If standard jobs are being used:

    - Identify appropriate standard jobs for each of the three components.
    - Identify appropriate standard jobs for the PM services.



This is not an exercise in AMT – use your pen and paper.

Notes

8 Creating Strategy Tasks

Procedure – Creating Strategy Tasks

You can create strategy tasks in many ways:

1. Under **Setup & Admin** > **Projections**, double-click **Strategy Task Manager** (or search **Strategy Task Manager**).

2. On the **Strategy Task Manager** form, click **Add** and select the **Type** of strategy task to create. See **RPM Training – AMT Key Concepts.**

3. Under **Setup & Admin** > **Projections – Multi-Editors**, double-click **Copy Strategy Tasks** (or search **Copy Strategy Tasks**).



*If **Update details of Strategy Tasks that already exist** is selected, additional options display.*

4. You can copy a projection, with all its strategy tasks, to an asset. Go to the main menu or the **Asset** form (**Base Details** tab) to copy the projection.



This feature mainly applies to copying a current projection (real asset) into a modelling scenario. It also applies to where you register an asset for the second time and you want to copy the strategy task last performed/next due details.

5. Import from the ERP system. See **AMT Interface specifications.**

6. From a standard job. See **RPM Training – AMT Standard Job.**

Exercise – Create Projections

1. Register six real assets and use the six strategy task creation methods outlined above to create current projections for each asset.

Exercise – Override Projections

1. Identify the most complete projection above and copy it to the other five assets to replace their current projection.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

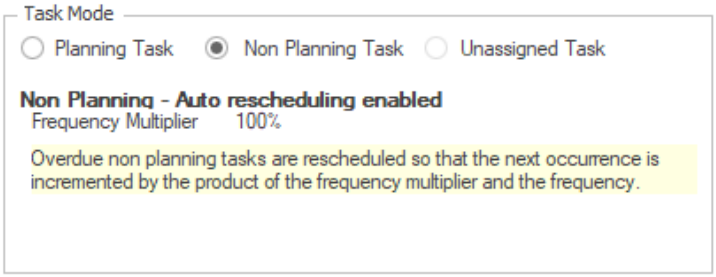
9 Completing Strategy Tasks

Overview

Depending on how an asset was created, some or all of the settings will be populated (that is, if you created a real asset from modelling).

Each tab in the form is explained below.

Base Details tab

Field	Description
Description	The description displays as the description on the work order, as well as some forms and reports. If you don't enter a description, it defaults to a concatenation of the four strategy task codes, for example: 1000:00:CC.0 – Engine
Task Codes	The combination of component code, modifier code, task type and task counter must be unique for a projection. **Tag Id** displays on Workscopes and work orders. It generally applies to fixed plant assets to help identify the correct asset when performing maintenance.
Task Mode	**Planning Task** – AMT automatically creates a work order when the planning lead time is reached (that is, when it is 33 days from the scheduled date). Planning tasks also appear in Long Term Planning and become overdue if they are not performed by the scheduled date. AMT life cycle costing assumes overdue components will be maintained today. **Non-Planning Tasks** are items you do not want to schedule, but want to manually create work orders when needed. If a non-planning task becomes overdue, AMT reschedules it for a percentage of its frequenc. **Frequency Multiplier** - this percentage is defined during the AMT configuration. For example, if a strategy task with a frequency of 10,000 Eng-Hrs becomes overdue and the frequency multiplier is 30%, it will be rescheduled for a further 3,000 Eng-Hrs.  **Unassigned Task** – these tasks are created by the system and cannot be edited. AMT will forecast a consistent daily cost for these items.

Costings tab

Field	Description
Categorisations	**Operational Criticality** – used for reference purposes only. **Manufacturer** – used for reference purposes only. **CBD** – used to flag strategy tasks that represent components with condition-based maintenance rules applied (only applicable for the Asset Ledger module).
Indexation	Select the applicable escalation category. See the **AMT Key Concepts – RPM Training Manual.**
Strategy Task Summary	Totals the costs/hours for all jobs. The job costs are adjusted for escalation and pricing adjustment factors.

Job Details tab

A strategy task can consist of multiple jobs.

The **Task Cost Summary** shows the sum of all the jobs with escalation and adjustment factors applied.

Field	Description
Job Header	**Cost Source** – identifies the source of the costs for the job. There are four sources: • **Manual** – enter costs manually. • **Standard Job** – select from a list of the AMT standard jobs. • **Cost Allocation** – create multiple cost records for a single job. Use multiple cost records in two situations: if multiple expense elements or cost centres apply or if you're interfacing with ERP maintenance plans or standard jobs. • **Consumables** – enter the consumable quantities and unit prices to calculate the parts cost. The job code identifies the type of consumable. **Cost Bearer** – a high-level cost responsibility grouping. See the **AMT Key Concepts – RPM Training Manual**. **Job Code** – identifies the type of job. **Currency** – mandatory field. **OH&S** – optional field, appears on the work order. **Library Link** – indicates if there is a document linked in the Library (it links if there is matching asset model + component code + task type + job code). **Cost Centre** – optional (note, if this field is left blank, when the work order is created, it defaults to the cost centre in the asset classifications). **Work Group** – optional. **Source of Supply** – optional, used by some inventory systems. **Commodity Code** – optional, only used in AMT Product Support Marketing (PSM) module. **BEC Code** – optional, only used in AMT PSM module.
Job Costs	**Pricing Date** is the date that the costs are applicable. For manually entered costs, escalation applies from the pricing date. The escalation schedule is defined in the **Base Details** tab under **Indexation**. For Current/Alternate/Centerline projections, the pricing date has no impact on jobs that are linked to standard jobs, as AMT assumes the pricing of the standard jobs is always up-to-date. **Expense Element** is an optional financial code, particularly used for financial budgeting and cost analyses.

Field	Description
Labour/Duration	**Labour Type** – this only impacts the forecast Maintenance Ratio report, which only includes the field labour hours in the calculation. **Labour Activity** – reflects the labour skill type. **Labour Hours** – direct labour hours to perform the job. **Duration** – is the elapsed time to perform the job.

Scheduling tab

This section contains the scheduling options for strategy tasks. The explanation of the different scheduling methods in AMT is covered in **AMT Scheduling Methods - RPM Training Manual.**

Field	Description
Scheduling Details	**Scheduling Method** – Date-based means scheduling in days. **UOM** - If scheduling is not date-based, you can selenct any UOM linked to that asset in a usage profile or its primary UOM as a scheduling method. **Schedule Type** – see the **AMT Scheduling Methods – RPM Training Manual**. **Not After** sets the end point when scheduling will stop. If it isn't selected, scheduling will continue to the end of the asset's term.
Last Change Prior To Equipment Start	These fields have NO IMPACT on scheduling. They are only used to determine the life that a strategy task achieved at its first actual occurrence. It is only applicable if used assets are registered in AMT.
Advanced Scheduling Logic	The default cost centre to apply to new strategy tasks and work orders created or imported without a cost centre.
External Linkage	These fields are used when integrating the strategy task to ERP maintenance plans. **External Identifier** - ERP maintenance plan.

Component tab

Field	Description
Parts Details	**Primary Part Number** – enables component forecasting by part number **Group Number** – a secondary parts reference number that enables searching for a group of parts in parts books. **Part Rating** – defines the size or capacity of a component (for example, 175kw for pumps). Only used for reporting and analytics. **Default Part Classification** – defines the type of component replacement strategy (Repair, Replace with new, Exchange, Reman, and so on). This field is primarily used for the Component Capacity Management module of AMT. In the **Next Due** tab, you can change the part classification for that next occurrence only. After that, it defaults to this part classification. Note that after the first changeout has occurred, the part classification entered here will also populate the **Next Due** tab as the default. **Default Location for Rebuild** - defines the default rebuild location of the component if it is to be rebuilt. This field is primarily used for forecasting the component repairs by the receiving rebuild location. In the **Next Due** tab, you can change the rebuild location for that next occurrence only. After that, it defaults to this rebuild location. After the first changeout has occurred, the rebuild location entered here will also populate the **Next Due** tab as the default.

Field	Description
	Starting Serial Number – used in the AMT Component Tracking module.
Warranty Period	This is an input into an alert that appears in the work order (**Feedback** tab). If this strategy task is performed for a second time within the tolerances set, a red flag appears on the work order, highlighting it is a potential warranty claim.

Strategy tab

This section is used to document the maintenance strategy and factor in contingency costs for component failures.

Field	Description
Performance Strategy	There are three performance strategies available: • On-Condition • On Schedule • Run To Failure If On-Condition is selected, you must select the component structure. The component structure is like a standard job for reliability rules and assumptions. In AMT, it can be used to factor in contingency costs and document the maintenance strategy around failure modes. It can also be used in the CBM module to store the condition monitoring interpretation rules. See **AMT Condition Based Maintenance – RPM Training Manual.**
View Details	If On-Condition and a component structure are selected and you click **View Details**, the form changes to the example below. This form shows: • **Analysis of Occurrence Modes** – the interpreted component structure • **Repair Reserve Task** – **A) System Calculation** -shows the calculations for the recommended repair reserve task for that risk profile. • **Repair Reserve Task** – **B) Task Details** – shows the repair reserve task that has been created, if there is one. • **Create RRT** button – you can create a repair reserve task if one has not been created. A repair reserve task is a strategy task specifically for factoring contingency costs.

Instructions tab

This section is used to link the strategy task to one of the following elements:

- Electronic workscopes.
- Linked documents, which can be any documents (PDF, Microsoft Word) in either the AMT Library or an external document management system.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

10 Strategy Task Adjustment Factors

Overview

A number of pricing adjustments can be applied when translating the strategy task job cost to the strategy task cost (see image below).

The **Audit Trail** form (accessed from **Quick Links**) provides the audit trail for the calculations.

The Projected Job - Audit Trail Details screen is displayed. The adjustments that can be applied are:

1. **Escalation** – escalation factors entered in the system table and applied from the pricing date in the strategy task job (see **AMT Key Concepts - RPM Training Manual**).
    - **Cost Adjustment Factors** – covers a range of items that can be entered in the **Audit Trail** form or the **Multi-Task Scope Editor**:
        - Parts Cost
        - Labour Split % (will only cost the percentage of labour entered)
        - Labour Cost
        - Labour Hours
        - Labour Preparation Time (will add the time at the start of every job)
        - Duration
        - Labour Rate
        - Crane (for crane hire)
        - Freight
        - Misc pricing
2. **Standard Job Pricing Adjustment Factors** – these adjustment factors are set in the Fleet system table editor. It applies adjustments to parts pricing only escalation.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

11 Editing Strategy Tasks

Exercise – Edit Strategy Tasks

1. Make the following changes to the five assets you created earlier:

    - Enter a labour share of 50% for PM services.
    - Apply a labour rate of $100 per hour to all strategy tasks.
    - Factor in a $200 crane hire for all component changeouts.
    - Enter a parts cost increase of 3% (as a factor, not one-off).
    - Run a report to forecast the tonnes moved and fuel litres for the next 5 years.
    - Link a strategy task to a standard job.
    - Link a set of job instructions to a strategy task.

Exercise – Forecast Consumables

2. Enter the following assumptions for engine oil and coolant to forecast their total life cycle consumption:

    - Engine oil – 50 litres every 250 Eng-Hrs at a cost of $1.20 per litre.
    - Coolant - 25 litres every 1,000 Eng-Hrs at a cost of $3.50 per litre.

Exercise – Factor Risk

3. Select a major component and factor in 20% after failure probability, at a random occurrence and at a cost of 25% greater than the before failure. Do this by creating a repair reserve task that is linked to a component structure.

12 Validate Strategy Tasks

Procedure – Validate Strategy Tasks

Perform the following steps to validate the strategy tasks for an equipment.

1. Double-click the equipment name from the **Equipment Navigator**.

   Real Equipment are identified by the  *icon.*

   The **Edit Equipment** dialog displays.

2. Click the **Tasks** tab and click **Apply Filter** to view the list of strategy tasks associated with the equipment.

   *By default, the **Projection** field is set to **Current**. The current projection has the live, up-to-date life cycle projection for the asset.*

3. The data displayed in the table is summarised at the bottom-right of the screen.

   - *The Total amount indicates the total life cycle cost of the asset, that is, actuals to date plus future.*
   - *Information on the asset's usage to date and future usage displays.*
   - *Information on the asset's cost per UOM also displays.*

4. Double-click a strategy task to open the **Edit Strategy Task** dialog.

   *By default, the **Base Details** tab is selected. This tab displays information such as component code, modifier code, task type, task counter and task mode*

5. Click the **Scheduling** tab and review the scheduling details:

    - Scheduling Method
    - Schedule Type
    - First Occurrence
    - Frequency
    - Advanced Scheduling Logic (for Planned Strategy Tasks)

6. Click the **Job Details** tab to check:

    - How the costs, labour hours and duration of the strategy task are defined.
    - Details of the cost bearer. Generally, the cost bearer is assumed to be **Maintenance**.



- *Cost Bearer is a filter on most of AMT's cost reports. Therefore, you should check before running the report whether to include or exclude Operations costs.*
- *If Cost Bearer is not a filter on a report, the report will include only maintenance costs because this is the default cost bearer in the system.*

7. If the **Cost Source** is **Standard Job**, you can also view the cost split derived from the linked standard job.

    - On the **Job Details** tab, click **Standard Job**.
    - The Review Cost Allocation dialog displays. Validate the costs.

8. To view the details of the standard job that the strategy task is linked to, click **Quick Links** on the **Job Details** tab.

9. Select **Edit Standard Job** from the list.

10. The Standard Job dialog displays. You can view basic information about the standard job by clicking on relevant tabs:

    - **Base Details:** The relevant codes and the pricing and cost details of the standard job.
    - **Operation:** A summary of the operations linked to the standard job.
    - **Parts:** Information on the parts that have come in with the operations.
    - **Labour:** Labour details.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

13 Create Additional Non-planning Strategy Tasks

Overview

When calculating the life cycle costing, you may need to include some costs for a more accurate calculation for the equipment. To account for these extra costs, you can create additional non-planning strategy tasks in AMT.

- *Do **NOT** use on real equipment in current projections.*
- *Before creating a non-planning strategy task in AMT, you must check and confirm with leaders.*

Procedure - Create Additional Non-planning Strategy Tasks

Perform the following steps to create additional non-planning strategy tasks.

1. Double-click the desired Real equipment name from the **Equipment Navigator**.
2. Click the **Tasks** tab and then click **Add** ().
3. The **Add Task** dialog displays. Select the required strategy task type:
    - Normal task
        - Can be a planning or a non-planning strategy task.
        - Select non-planning strategy task.
        - Can be linked to standard jobs.
        - Can use advanced scheduling logic.
    - Unassigned task
        - Cannot use advanced scheduling logic.
        - Cannot be linked to a standard job.
    - Other Operating Cost task
    - Capital tasks
4. Click **OK**.
5. The **Add Strategy Task** dialog displays. Enter the following details for a non-planning strategy task:
    - **Description**: The description for the new task.
    - **Component Code**: Code for the component you are going to work on. For example: Engine.
    - **Modifier Code**: To be used if you need to differentiate between areas. For example: Front or Rear.
    - **Task Type**: Defined by the type of work. For example: Schedule component change.
    - **Task Counter**: Set to **0 – (None)** unless you have the same component code, modifier code and task type.
6. In the **Task Mode** section, select **Non-Planning Task**.
7. Select the **Requires STO** check box if relevant.

8. Click the **Scheduling** tab to add the scheduling method:
    - Set the **Scheduling Method** as **Date Based** or **Usage Based**.
        - If **Usage Based** is selected, select the relevant UOM. Note that only UOMs linked to the equipment are available for selection.
    - Set the **Schedule Type** as **Last Performed** or **Last Scheduled**.
    - Set the **First Occurrence**.
    - Set the **Frequency**.
    - If required, select the **Not After** check box to prevent AMT from forecasting this task beyond the specified date or usage.
    - Click **Save**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

14 Set Frequency of Non-planning Strategy Tasks

Overview

An estimated frequency is set for non-planning strategy tasks so that AMT can forecast these details. If a non-planning strategy task becomes overdue, AMT reschedules it for a predefined percentage of its frequency. This percentage is set to 33%.

The non-planning strategy tasks are identified in AMT in the **Edit Strategy Task** tab of the **Equipment Editor** screen if the check box in the **P** column (P: Planned) is clear.

Procedure – Set Frequency of Non-planning Strategy Tasks

Perform the following steps to set the frequency of non-planning strategy tasks:

1. Open the **Edit Strategy Task** dialog for a desired non-planning strategy task.



When reconsidering the frequency of a non-planning task, ask yourself, "How often will this task be performed over the asset's life cycle?" This should be done in conjunction with the Reliability & Maintenance engineering teams. Considerations include:

- *Component age*
- *Average historical frequency of task*
- *Upcoming usage requirements*
- *Asset health*

2. Click the **Scheduling** tab. Perform the following actions:

- Update the **First Occurrence** and **Frequency**.
- Set the **Scheduling Method** as **Date Based** or **Usage Based**.
- Enter an appropriate value in the **First Occurrence** field. If it is usage-based, enter the relevant hours; if it is date-based, enter the relevant number of days or the specific date.
- In the **Frequency** field, set the frequency you anticipate for the task.
- Click **Save**.



This step will change the frequency of the non-planning strategy task, influence the forecasting in the future and help calculate a more accurate life cycle cost of the asset.

- Click **Apply Filter** on the **Edit Equipment** dialog. A list of updated tasks display.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

15 Create/Update Unassigned Tasks

Overview

Unassigned tasks account for additional costs that have not already been covered by a planning or non-planning strategy task. These are created or updated so that the life cycle costing is accurate.



Unlike planning and non-planning strategy tasks, AMT forecasts the costs of unassigned tasks based on the unassigned task's costs, frequency, and the projected days or usage per month (depending whether the unassigned task is based on usages or days).

Procedure - Create/update Unassigned Tasks

Perform the following steps to create or update unassigned tasks.

1. Double-click the desired real equipment from the **Equipment Navigator**. The **Edit Equipment** dialog displays.
2. Click the **Tasks** tab and then click **Add** ().
3. The **Add Task** dialog displays. Select **Unassigned Task** and then click **Search**.
4. Select the appropriate subsystem from the list and click **OK**.
5. Next, click **OK** on the **Add Task** dialog.



Unassigned tasks are not at the component level. These costs account for general work on a particular area of an equipment – referred to as a subsystem.

6. The **Task Mode** indicates that this task is Unassigned.



The Component Code, Modifier Code, Task Type and Task Counter codes are predefined and cannot be changed.

7. Click the **Job Details** tab. Click the **Cost Source** field. In the **Result List** dialog, select **Manual**.
8. By default, the cost fields for an unassigned task are zero. Enter appropriate values for parts, labour, miscellaneous and other options in the relevant fields in the **Job Costs** section, such as **Expense Elements**.

9. If more than one expense element is required for parts or labour, you need to create a **Cost Allocation**. Click in the **Cost Source** field and choose the **Cost Allocation** option.

10. Click **Add**.

11. The **Add Cost Allocation** dialog displays. Perform the following actions:

    - Click **Add** to add new costs. You can add or modify expense elements.
    - Click **Save** when done.

12. The changes are reflected in the **Job Costs** section. Click **Save**.

13. Enter appropriate scheduling details, such as **Scheduling Method** and **Frequency**. Click the **Scheduling** tab.

14. Click **Save** when finished.

16 Review and Validate the Strategy

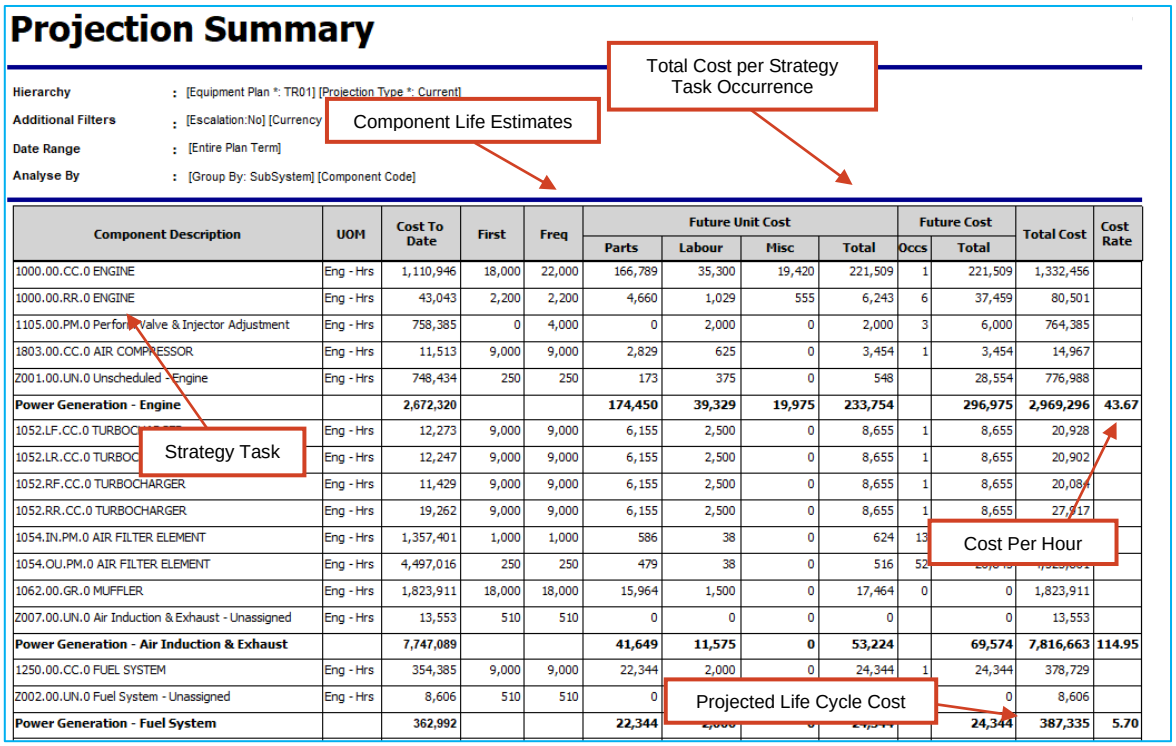
Overview

The final step is to review and validate the maintenance strategy. While considering life cycle costing, you should analyse whether the outputs meet the required performance parameters and align with expectations (OEM quotes, history with similar equipment, and so on).

After the projections are created, validate the costs before the Equipment, Plant, or Asset Manager signs off.

To validate the costs in the projections:

1. Run the **Projection Summary** report.
   - Select **Projection Type**.
   - Set **Escalation** = **No**.



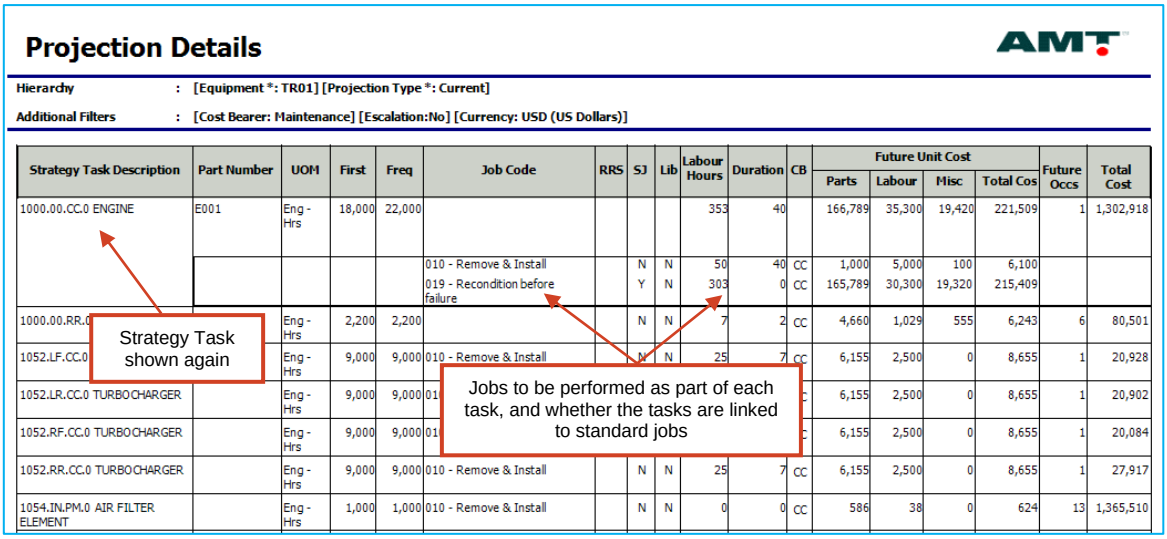
Component Description	UOM	Cost To Date	First	Freq	Future Unit Cost				Future Cost		Total Cost	Cost Rate
					Parts	Labour	Misc	Total	Occs	Total		
1000.00.CC.0 ENGINE	Eng - Hrs	1,110,946	18,000	22,000	166,789	35,300	19,420	221,509	1	221,509	1,332,456	
1000.00.RR.0 ENGINE	Eng - Hrs	43,043	2,200	2,200	4,660	1,029	555	6,243	6	37,459	80,501	
1105.00.PM.0 Perform Valve & Injector Adjustment	Eng - Hrs	758,385	0	4,000	0	2,000	0	2,000	3	6,000	764,385	
1803.00.CC.0 AIR COMPRESSOR	Eng - Hrs	11,513	9,000	9,000	2,829	625	0	3,454	1	3,454	14,967	
Z001.00.UN.0 Unscheduled - Engine	Eng - Hrs	748,434	250	250	173	375	0	548		28,554	776,988	
Power Generation - Engine		**2,672,320**			**174,450**	**39,329**	**19,975**	**233,754**		**296,975**	**2,969,296**	**43.67**
1052.LF.CC.0 TURBOC	Eng - Hrs	12,273	9,000	9,000	6,155	2,500	0	8,655	1	8,655	20,928	
1052.LR.CC.0 TURBOC	Eng - Hrs	12,247	9,000	9,000	6,155	2,500	0	8,655	1	8,655	20,902	
1052.RF.CC.0 TURBOCHARGER	Eng - Hrs	11,429	9,000	9,000	6,155	2,500	0	8,655	1	8,655	20,084	
1052.RR.CC.0 TURBOCHARGER	Eng - Hrs	19,262	9,000	9,000	6,155	2,500	0	8,655	1	8,655	27,917	
1054.IN.PM.0 AIR FILTER ELEMENT	Eng - Hrs	1,357,401	1,000	1,000	586	38	0	624	13			
1054.OU.PM.0 AIR FILTER ELEMENT	Eng - Hrs	4,497,016	250	250	479	38	0	516	52			
1062.00.GR.0 MUFFLER	Eng - Hrs	1,823,911	18,000	18,000	15,964	1,500	0	17,464	0	0	1,823,911	
Z007.00.UN.0 Air Induction & Exhaust - Unassigned	Eng - Hrs	13,553	510	510	0	0	0	0		0	13,553	
Power Generation - Air Induction & Exhaust		**7,747,089**			**41,649**	**11,575**	**0**	**53,224**		**69,574**	**7,816,663**	**114.95**
1250.00.CC.0 FUEL SYSTEM	Eng - Hrs	354,385	9,000	9,000	22,344	2,000	0	24,344	1	24,344	378,729	
Z002.00.UN.0 Fuel System - Unassigned	Eng - Hrs	8,606	510	510	0					0	8,606	
Power Generation - Fuel System		**362,992**			**22,344**					**24,344**	**387,335**	**5.70**

1. Are all the strategy tasks set up with the correct codes?
2. Are the estimated component lives accurate based on the operating environment, historic performance, and so on?
3. Has sufficient provision been built into the life cycle for minor repairs? (that is, have unassigned tasks been created in the relevant subsystems with budgets?)
4. Is the cost per hour reasonable?

Additional details are in the following reports.

Projection Details

Shows additional details of each strategy task for the selected projection.



Projection Details

AMT

Hierarchy : [Equipment *: TR01] [Projection Type *: Current]

Additional Filters : [Cost Bearer: Maintenance] [Escalation:No] [Currency: USD (US Dollars)]

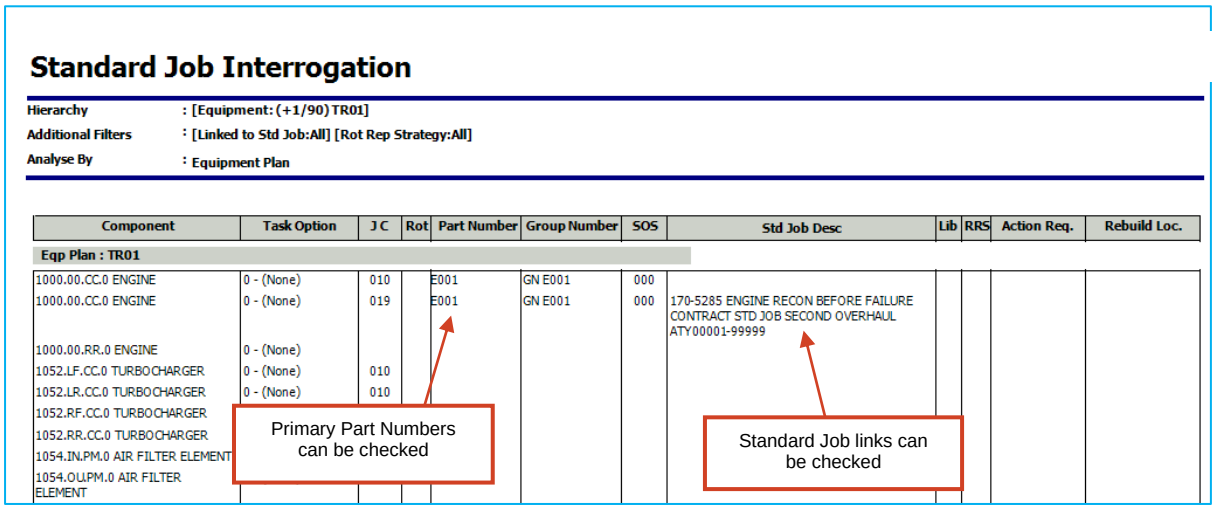
Strategy Task Description	Part Number	UOM	First	Freq	Job Code	RRS	SJ	Lib	Labour Hours	Duration	CB	Future Unit Cost				Future Occs	Total Cost
												Parts	Labour	Misc	Total Cos		
1000.00.CC.0 ENGINE	E001	Eng - Hrs	18,000	22,000					353	40		166,789	35,300	19,420	221,509	1	1,302,918
					010 - Remove & Install		N	N	50	40	CC	1,000	5,000	100	6,100		
					019 - Recondition before failure		Y	N	303	0	CC	165,789	30,300	19,320	215,409		
1000.00.RR.0		Eng - Hrs	2,200	2,200			N	N	7	2	CC	4,660	1,029	555	6,243	6	80,501
1052.LF.CC.0		Eng - Hrs	9,000	9,000	010 - Remove & Install		N	N	25	7	CC	6,155	2,500	0	8,655	1	20,928
1052.LR.CC.0 TURBOCHARGER		Eng - Hrs	9,000	9,000	01						C	6,155	2,500	0	8,655	1	20,902
1052.RF.CC.0 TURBOCHARGER		Eng - Hrs	9,000	9,000	01						C	6,155	2,500	0	8,655	1	20,084
1052.RR.CC.0 TURBOCHARGER		Eng - Hrs	9,000	9,000	010 - Remove & Install		N	N	25	7	CC	6,155	2,500	0	8,655	1	27,917
1054.IN.PM.0 AIR FILTER ELEMENT		Eng - Hrs	1,000	1,000	010 - Remove & Install		N	N	0	0	CC	586	38	0	624	13	1,365,510

Strategy Task shown again

Jobs to be performed as part of each task, and whether the tasks are linked to standard jobs

Standard Job Interrogation

This report should only be used if strategy tasks were linked to standard jobs or were linked to primary part numbers.



Standard Job Interrogation

Hierarchy : [Equipment: (+1/90) TR01]

Additional Filters : [Linked to Std Job:All] [Rot Rep Strategy:All]

Analyse By : Equipment Plan

Component	Task Option	J C	Rot	Part Number	Group Number	SOS	Std Job Desc	Lib	RRS	Action Req.	Rebuild Loc.
Eqp Plan : TR01											
1000.00.CC.0 ENGINE	0 - (None)	010		E001	GN E001	000					
1000.00.CC.0 ENGINE	0 - (None)	019		E001	GN E001	000	170-5285 ENGINE RECON BEFORE FAILURE CONTRACT STD JOB SECOND OVERHAUL ATY00001-99999				
1000.00.RR.0 ENGINE	0 - (None)										
1052.LF.CC.0 TURBOCHARGER	0 - (None)	010									
1052.LR.CC.0 TURBOCHARGER	0 - (None)	010									
1052.RF.CC.0 TURBOCHARGER											
1052.RR.CC.0 TURBOCHARGER											
1054.IN.PM.0 AIR FILTER ELEMENT											
1054.OU.PM.0 AIR FILTER ELEMENT											

Primary Part Numbers can be checked

Standard Job links can be checked

- When the asset has been set up in AMT properly, provide the **Projection Summary**, **Projection Details** and **Projection and Interval Analysis** report to the Equipment, Plant, or Asset Manager for their approval.
- The **Maintenance Strategy Summary** report can also provide a useful record of the strategy.

Additional reports that are useful for reviewing and validating the strategy include:

Reports	Reviewing	Validating
Productivity	Forecast availability Forecast utilisation	**Forecast Availability** **Usage Analysis Chart/Details**
Costs	Total costs Cost per Unit of Measure % Unscheduled costs	**Cost Analysis – "Life-cycle"** **Cost per UOM** **Cost Analysis – "Life-cycle" (Task Type)**
Labour	Total labour hours by skill type Labour hours to operating hours	**Labour Hours forecast** **Maintenance Ratio**
Resources	Parts/component inventory required	**Component Forecast**
Consumables	Consumable usage	**Consumables Forecast**
Economic life	Analyse value in components over time	**RVIC Chart**

Exercise – Review and Validate the Strategy

1. Use the reports listed in this section to validate the maintenance strategy you created.
2. Understand the output of each report.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

17 Editing Multiple Strategy Tasks

Overview

There are multiple ways to add, edit, and delete strategy tasks in the projections of an asset. The appropriate method depends on the types of changes required.

When editing the equipment using the **Equipment Plan Editor**, there is a tab called **Quick Editor**. You can change certain details of multiple strategy tasks within the same equipment.

Other multi -ditors exist within AMT. A brief description of each editor is provided below, along with an example of when they are useful.

Type of Edit Required	AMT Function
Change a small number of selected tasks.	Strategy Task Manager.
Change costs by a nominated factor across multiple tasks.	Multi-Task % Cost Adjuster.
Adjust parts prices on linked standard jobs.	Find the relevant fleet using the **Fleet Administration** function under which the asset is located; edit and apply the price factors.
Change costs across multiple equipment in the same fleet.	Multi-Task Editor.
Change the characteristics of a strategy task across multiple assets.	Multi-Task Editor. Multi-Task Labour Editor. Multi-Task Scope Editor.
Copy a strategy task to another projection.	Copy New Projected Task.
Delete a task from many equipment.	Multiple Equip Delete Task.
Update forecast costs based on actual costs.	Forecast Cost Adjuster.

For more information on the multi-editors, refer to the **AMT Strategy Optimisation Training Manual**.

18 Escalation

Overview

Escalation refers to price changes over time; sometimes it's also called indexation. For example, if the initial projection is priced in year 1, for a life cycle spanning several years, then actual costs could be higher than the initial pricing, simply as a function of inflation over time. To harmonise comparisons between projections, apply escalation to budget prices to adjust these values to a current level.



IMPORTANT: Future projections of both current and budget projections are made at today's escalated values.

Summary of Cost Escalation Rules in AMT

Projection Type	Actual Costs	Future Costs
Budget	Not Escalated	Not Escalated
Current and Alternate	Not Escalated	Escalated to today's value
Archived	Not Escalated	Escalated to the point of archiving

For more information on escalation in AMT, refer to the **AMT System Administration Training Manual**.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

19 Summary

Summary

The key points to remember from this guide are:

- Strategy definition is the process of validating the maintenance strategy in AMT.
- You can perform the following strategy definition steps in AMT:
    - Decide Base Operations Parameters
    - Consider Asset Capability
    - Define Application Severity
    - Define Asset Structure
    - Decide Maintenance Strategy
    - Determine Task Scheduling
    - Define Repair Strategy and Resources
    - Creating Strategy Tasks
    - Completing Strategy Tasks
    - Strategy Tasks Adjustments Factors
    - Editing Strategy Tasks
    - Validate Strategy Tasks
    - Create additional Non-planning Strategy Tasks
    - Set frequency of Non-planning Strategy Tasks
    - Create/update Unassigned Tasks
    - Review and Validate the Strategy
    - Editing Multiple Strategy Tasks
    - Escalationa